Magic 3D Studio: From A Single Image To Beautiful 3D Scene & Photos

Two-agent pipeline on **AWS Lambda** with validated, cookbook-driven generation

1 MOBILE APP (React Native / Expo)

Customer photo
The customer's picture always stays a picture and is never rebuilt in 3D.

Magic3DStudio app

Live 3D preview (WebView)

Record / Share

Retouch button on the canvas (beside Record)

Library (expo-sqlite)
stores the final HTML, the scene images and the product-view images, so any scene can be reopened and retouched offline.

Post-processing (glow, optional toggle in Settings)

2 AWS (Lambda orchestrator)

AWS Lambda (Java)

Config and ReferenceStore
Reference themes (Train, Ocean, Portal and more) with style guidance.

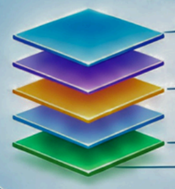
Prompts
CINEMATIC TOOLBOX (36 techniques) + PRODUCT-TYPE RECIPES.

LimClient
The model router.

Validator
SceneChecker (static and syntax checks) and SceneAuditor (quality audit).

MagicHtmlAssembler
Merges shell.html + cookbook.js + brief + scene into one HTML page.

One portable file



- **shell.html** (stage, camera, renderer, UI)
- **cookbook.js** (ready-made, tested 3D lighting blocks, additive-only, API-indexed)
- **brief (JSON)**
- **scene code**

3 THE TWO AGENTS

Agent A Art Director
Input: photo + user prompt + reference theme.
Output: BRIEF: concept, palette, camera plan, mood, key objects, motion beats, product-type recipe.

brief

Agent B 3D Engineer
Input: brief + the cookbook API.
Output: buildScene(), brief() JavaScript that composes cookbook functions.

The Cookbook
lighting · materials · particles · camera rigs · live controls (sliders, HUD, modes) · x-ray / assembly · glow

- Separation of creative vs technical work
- Constrained by a tested cookbook, so fewer broken scenes
- Reference themes give consistent, premium styling.

4 VALIDATION AND TWO STAGES

auto-repair: errors and audit notes go back to Agent B (retry)

Retouch (user instruction)

Stage 1: Build
Agent B writes the full scene.

Stage 2: Dress (Advanced mode)
dressScene adds polish, props, atmosphere and lighting.

Validator: SceneChecker + SceneAuditor
Safe check, audit composition, motion and product.

Safe fallback scene

v1 · v2 · v3

5 CLOUD MODEL PROVIDERS

Image generation
Meta (scene image and product view)

3D Scene generation (Load-balanced)

Load Balancer (auto distribute)

Modal · **Meta**

Generate 3D Scene (via Modal or Meta)

LimClient routing: load-balance + fallover between Modal and Meta for 3D scene generation

Provider-agnostic: swap or add models without changing the app.

HOW WE SCALE QUALITY FURTHER

- **Grow the cookbook** (new tested building blocks, additive only).
- More reference themes and product-type recipes.
- **Stronger SceneAuditor:** automatic screenshot scoring (render, then a vision model critiques).
- **Best-of-N:** generate several candidate scenes in parallel across Modal and Meta.
- Post-processing (glow, with more effects) and device-adaptive quality with an automatic fps fallback.
- **Feedback loop:** user Retouch and Library favorites train better prompts and recipes.
- (Optional, future) higher-fidelity rendering such as WebGPU.

= Main flow · = Fallback / optional · = Retouch loop · = Validation